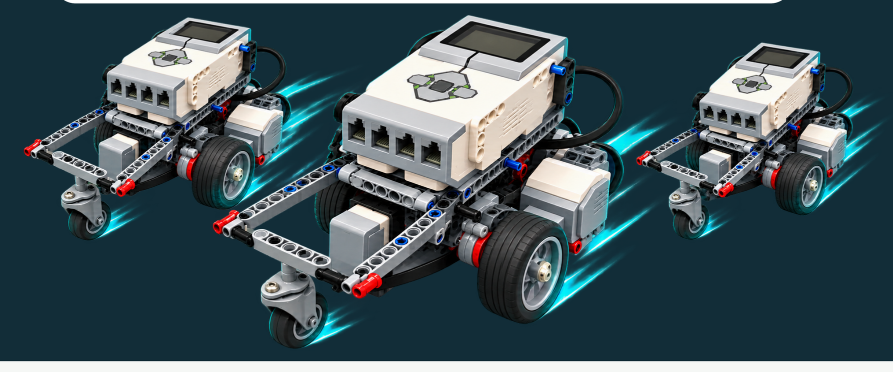
MINDSTORMS PARTY

UNIVERSITY OF BAYREUTH / BACHELOR PROJECT

Robot Control & Communication

Collision avoidance, steering & network protocols

UNIVERSITÄT BAYREUTH



01 One system. Three communication channels.

ARCHITECTURE

DIRECT PHONE CONTROL / UDP 7778

AR App
Unity / C#
Joystick + game interface

TCP 1234 — game events

Desktop
C++ / Qt / OpenCV
Camera, planner, path follower

UDP 7778 — autonomous drive — UDP 7779

EV3 robot
Native Rust / ev3dev
Motor execution + status

UDP broadcast: device discovery on the local network

1 Hz heartbeats report state and refresh robot IPs.

3 s without status: desktop marks it offline.

Sends: `mind/<robot_id>/cmd`
Receives: `mind/+/status`

MOSQUITTO / MQTT 1883
Auto reconnect + restore subscriptions

Receives: `mind/<robot_id>/cmd`
Sends: `mind/<robot_id>/status`

UDP: binary for low latency

4-byte binary packets cut message size and parsing time for **low latency**. Default: **20 Hz**. New setpoints replace old ones without retries.

01	01	32	00
ev3_01	drive	+50%	0% turn

Binary drive packet: 4 bytes, shown in hex

TCP: structured game events

Phones exchange **JSON game events** with the desktop. The **type** field selects a handler, e.g. **shop_buy**. Shared game updates reach all connected phones; **shop_open** carries the player's shop inventory.

Splatoon UDP: input → desktop **12346**; grid updates → phones **12345**.

MQTT: commands & replies

Binary **drive_for, turn, stop, ping** use **QoS 1 (at least once)**; ACKs echo IDs. Both clients restore subscriptions. **Desktop:** automatically reconnects, with up to 10 seconds between retry attempts.

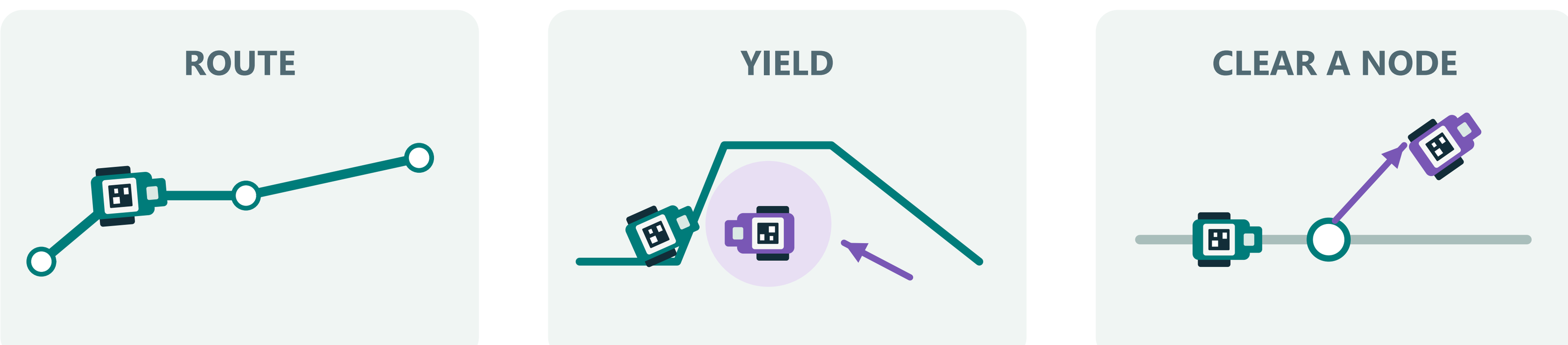
`mind/<robot_id>/cmd`
`mind/<robot_id>/status`

02 From a target to controlled movement

DESKTOP CONTROL

Plan around other robots

Global board routes + local collision avoidance



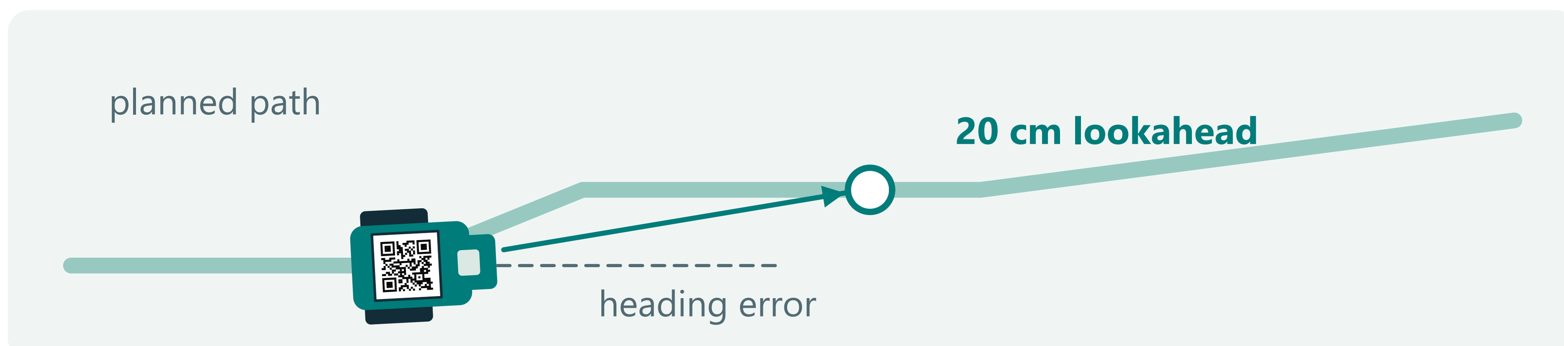
1 / Route. Dijkstra selects the board-node route. Grid A* detours around parked robots; clearance-checked corner rounding smooths the waypoints.

2 / Yield. A central coordinator watches nearby paths. The higher robot ID yields, replans around the other robot, or holds if no detour exists.

3 / Clear. If a destination is occupied, the blocker is routed to a free parking spot off the lanes. The arriving robot waits for the node to clear.

Follow the path, correct the heading

Camera pose → lookahead target → steering → wheels



1 / Observe. ArUco markers provide position and heading. A board homography converts camera pixels into physical centimetres.

2 / Steer. A lookahead point sets the heading error. A PID controller computes turn rate; forward speed falls in corners and near the goal.

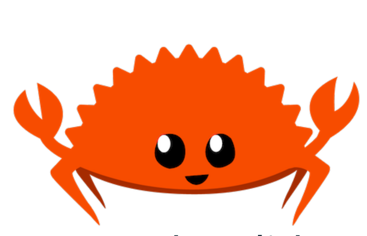
3 / Settle. Sharp turns use pivot pulses and pauses for camera feedback to catch up. Calibration scales compensate drive and turn bias.

AUTONOMOUS DRIVING — **Plan the route** → **Avoid obstacles** → **Steer with camera feedback**

03 Inside the brick

RUST + EV3DEV

Ferris at the wheel



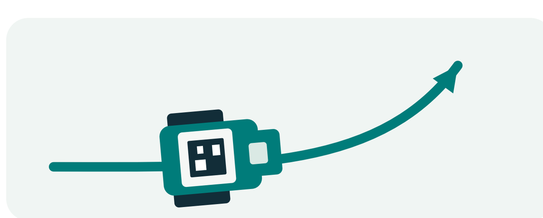
Four workers, one motor controller

Native Rust runs directly on the EV3's Linux system. A mutex protects shared drive state; each valid UDP packet overwrites the setpoint and wakes the motor worker. A 10 ms timeout keeps watchdog checks running between packets.

UDP	MOTOR	MQTT	HEARTBEAT
Decode + ID filter Wake the motor loop	10 ms check interval Regulated B/C motors	Dispatch commands Interruptible moves	1 Hz to desktop State + battery + ID

Motion commands: stream speed + turn for continuous driving; queue **drive_for** and **turn** for distance and angle moves. The motor worker sends a completion ACK for queued moves.

Differential drive: $v_L = v - \omega b/2$; $v_R = v + \omega b/2$.
v: forward speed; ω: turn rate; b: wheel spacing.
At the speed limit, both wheels slow together to keep the curve.
Motors B/C; 56 mm wheels; b = 126 mm.



Independent stop mechanisms

150 ms	No fresh drive packet: the EV3 watchdog stops continuous motion.
400 ms	Stale camera pose: the desktop commands zero drive.
STOP	MQTT sets an interrupt flag for precision moves; motors poll it.

MORE ABOUT THE PROJECT

Scan for project documentation →